



COMSATS UNIVERSITY ISLAMABAD (CUI)

DEPARTMENT OF COMPUTER SCIENCE
TERMINAL EXAMINATION SPRING - 2026
BS (SE)– VI SEMESTER

Course: AIC354 - Machine Learning Fundamentals Lab

Maximum Marks: 50

Dated: May 17, 2026

Instructor: Mr. Anayat Ullah

Due Date: May 24, 2026

“An End-to-End Machine Learning Framework for House Price Prediction”

Project Objective

The objective of this project is to develop a House Price Prediction System using real-world property listings collected from Zameen.com.

Students will be required to:

- Scrape property listing data from the website
- Store the collected data in CSV format
- Perform data preprocessing
- Train machine learning regression models
- Predict house prices based on property features
- Compare model performance

(CLO – 1: Data Acquisition & Web Scraping)

Task 1: Data Collection

Students are required to scrape 300–400 property listings from Zameen.com for the city of Islamabad. The dataset should include the following features:

- Price (target variable)
- Area
- City
- Number of bedrooms
- Number of bathrooms
- Location
- Property type
- Built in year
- Parking space
- Servant Quarters
- Store rooms
- Kitchens
- Drawing Rooms, etc.

(CLO – 2: Data Preprocessing & Feature Engineering)

Task 2: Data Preprocessing

Students must preprocess the collected dataset by performing the following tasks:

- Handle missing values
- Remove duplicate records
- Convert categorical variables (e.g., location, property type) into numerical form
- Perform train-test split

(CLO – 3: Machine Learning Model Development)

Task 3: Machine Learning Model Development

Students are required to implement the following algorithms:

- Linear Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost
- CatBoost

(CLO – 4: Model Evaluation & Performance Analysis)

Task 4: Model Evaluation

Evaluate the performance of all models using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Students must compare all models and identify the best-performing model.

(CLO – 5: End-to-End System Development)

Task 5: Final System

Students should develop a simple prediction system where a user enters:

- Area
- Number of bedrooms
- Number of bathrooms
- Location

The system should output: ***Estimated House Price***

Deliverables

Students must submit the following in a ZIP file:

1. Python scraper code
2. CSV dataset
3. Machine learning training code
4. Final report

Final Report Must Include

- Problem statement
- Dataset description
- Machine learning algorithms used
- Results and performance comparison
- Challenges faced during implementation

Bonus Marks

Additional marks will be awarded to students who implement:

- Better scraper automation
- Improved model accuracy

Project Overview

In this project, students are required to develop a machine learning-based system that predicts house prices using real estate listings collected from Zameen.com.

Students will first collect their own dataset by developing a web scraper. The scraper will extract real estate listing information from the website and store it in a structured dataset. This dataset will then be used to train machine learning models for predicting house prices.

The project is intended to expose students to real-world machine learning workflows involving:

- Data collection
- Data preprocessing
- Feature engineering
- Model training
- Model evaluation
- Prediction system development

Dataset Requirements

Students must create their own dataset by scraping listings only from the city of Islamabad. The dataset should contain at least 300 to 400 property listings.

The final dataset should include the following attributes:

- Price (target variable)
- Area
- City
- Number of bedrooms
- Number of bathrooms
- Location
- Property type
- Built in year
- Parking space
- Servant Quarters
- Store rooms
- Kitchens
- Drawing Rooms, etc.

Web Scraper Requirements

Students must develop a Python-based web scraper capable of collecting property listing information from Zameen.com.

The scraper should be able to:

- Access listing pages
- Extract required information
- Handle multiple property listings
- Navigate through multiple pages
- Store extracted data in CSV format

Students may use:

- Python, BeautifulSoup, Requests, Selenium, etc.

Data Preprocessing Requirements

After collecting the dataset, students must preprocess the data before applying machine learning algorithms.

This includes:

- Handling missing values
- Removing duplicate records
- Converting categorical variables into numerical form
- Cleaning inconsistent formats
- Feature selection
- Splitting dataset into training and testing sets

Machine Learning Requirements

Students are required to implement regression algorithms to predict house prices.

Recommended algorithms include:

- Linear Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost
- CatBoost

Model Evaluation

Students must evaluate model performance using appropriate regression metrics such as:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Students should compare model performance and justify which model performs best.

Final System

The final output should be a working prediction system where a user can input house features such as:

- Area
- Bedrooms
- Bathrooms
- Location

The system should return the predicted house price.